In [1]:
```
import numpy as np    # array
```

In [2]:
```
import matplotlib.pyplot as plt    # Visualization
```

In [4]:
```
from PIL import Image    # python image library
```

In [12]:
```
im_image = Image.open(r'C:\Users\HP\OneDrive\Documents\Full Stack Data Science W
```

In [13]:
```
im_image
```

Out[13]:

In [15]:
```
type(im_image)
```

Out[15]:
```
PIL.JpegImagePlugin.JpegImageFile
```

In [19]:
```
image_array =np.asarray(im_image)
```

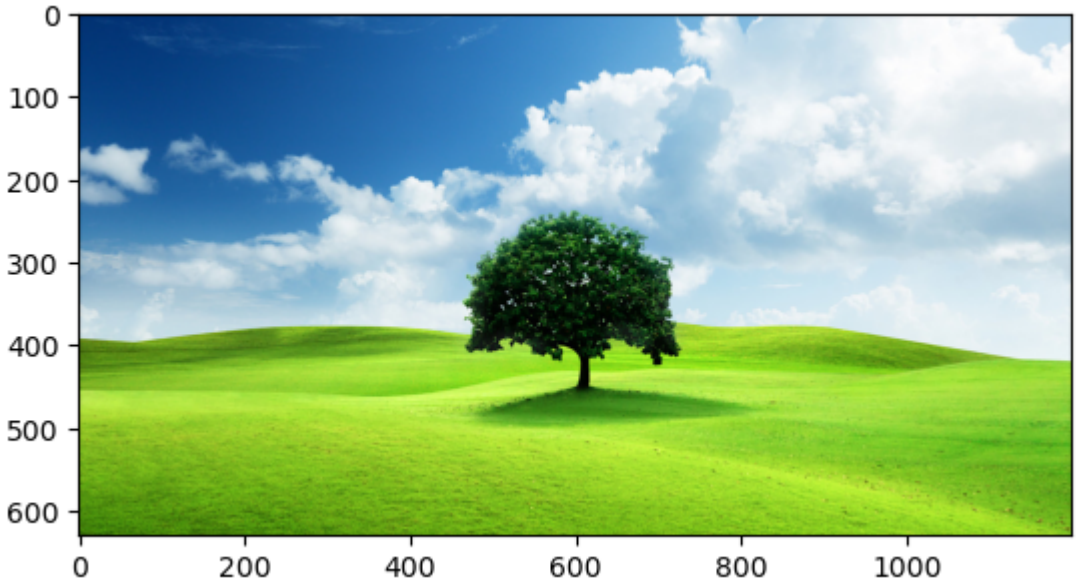
In [20]:
```
image_array
```

Out[20]:
```
array([[[  1,  49, 113],
        [  2,  50, 114],
        [  4,  49, 114],
        ...,
        [193, 220, 237],
        [194, 221, 238],
        [194, 221, 238]],

       [[  1,  49, 113],
        [  2,  50, 114],
        [  4,  49, 114],
        ...,
        [194, 221, 238],
        [194, 221, 238],
        [194, 221, 238]],

       [[  1,  49, 113],
        [  2,  50, 114],
        [  4,  49, 114],
        ...,
        [194, 221, 238],
        [194, 221, 238],
        [194, 221, 238]],

       ...,

       [[ 65, 123,  10],
        [ 68, 128,   6],
        [ 67, 124,  19],
        ...,
        [126, 194,  33],
        [121, 192,  26],
        [124, 198,  27]],

       [[ 65, 124,   8],
        [ 62, 119,   2],
        [ 60, 115,  14],
        ...,
        [139, 206,  41],
        [130, 199,  30],
        [133, 205,  33]],

       [[ 61, 120,   4],
        [ 54, 110,   0],
        [ 42,  96,   0],
        ...,
        [144, 209,  43],
        [135, 202,  34],
        [134, 202,  31]]], shape=(630, 1200, 3), dtype=uint8)
```

In [21]:
```
image_array.shape
```

Out[21]:
```
(630, 1200, 3)
```

In [23]:
```
plt.imshow(image_array)
plt.show()
```



In [24]:
```
type(image_array)
```

Out[24]:
```
numpy.ndarray
```

In [26]:
```
image_red = image_array.copy()
```

In [27]:
```
image_red
```

Out[27]:
```
array([[[  1,  49, 113],
        [  2,  50, 114],
        [  4,  49, 114],
        ...,
        [193, 220, 237],
        [194, 221, 238],
        [194, 221, 238]],

       [[  1,  49, 113],
        [  2,  50, 114],
        [  4,  49, 114],
        ...,
        [194, 221, 238],
        [194, 221, 238],
        [194, 221, 238]],

       [[  1,  49, 113],
        [  2,  50, 114],
        [  4,  49, 114],
        ...,
        [194, 221, 238],
        [194, 221, 238],
        [194, 221, 238]],

       ...,

       [[ 65, 123,  10],
        [ 68, 128,   6],
        [ 67, 124,  19],
        ...,
        [126, 194,  33],
        [121, 192,  26],
        [124, 198,  27]],

       [[ 65, 124,   8],
        [ 62, 119,   2],
        [ 60, 115,  14],
        ...,
        [139, 206,  41],
        [130, 199,  30],
        [133, 205,  33]],

       [[ 61, 120,   4],
        [ 54, 110,   0],
        [ 42,  96,   0],
        ...,
        [144, 209,  43],
        [135, 202,  34],
        [134, 202,  31]]], shape=(630, 1200, 3), dtype=uint8)
```

In [30]:
```
image_array==image_red
```

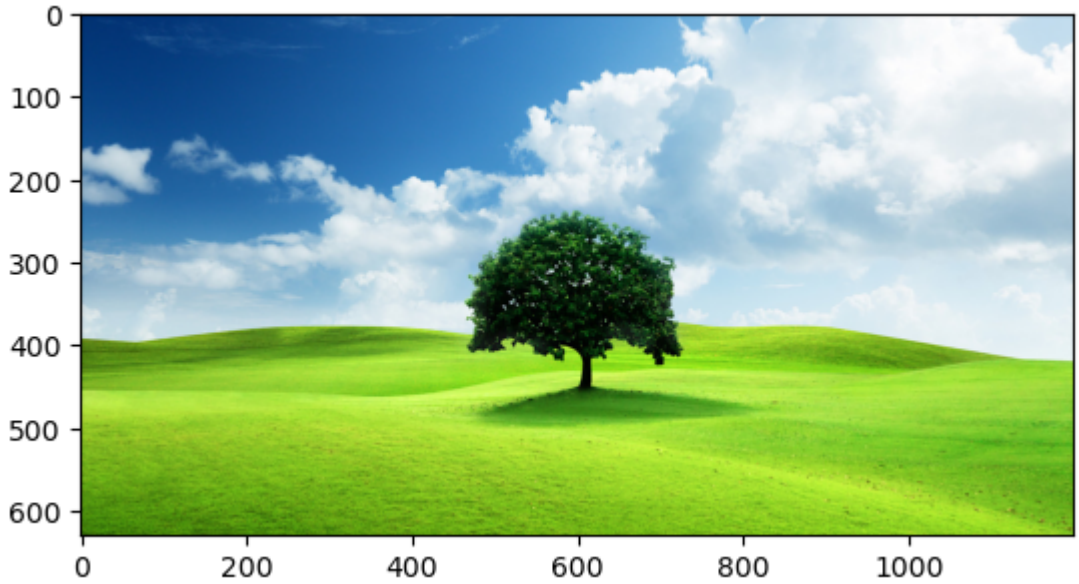
Out[30]:
```
array([[[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       ...,

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]]], shape=(630, 1200, 3))
```

In [31]:
```
type(im_image)
```

Out[31]:
```
PIL.JpegImagePlugin.JpegImageFile
```

In [32]:
```
plt.imshow(image_red)
```

Out[32]:
```
<matplotlib.image.AxesImage at 0x21d0f9b5810>
```

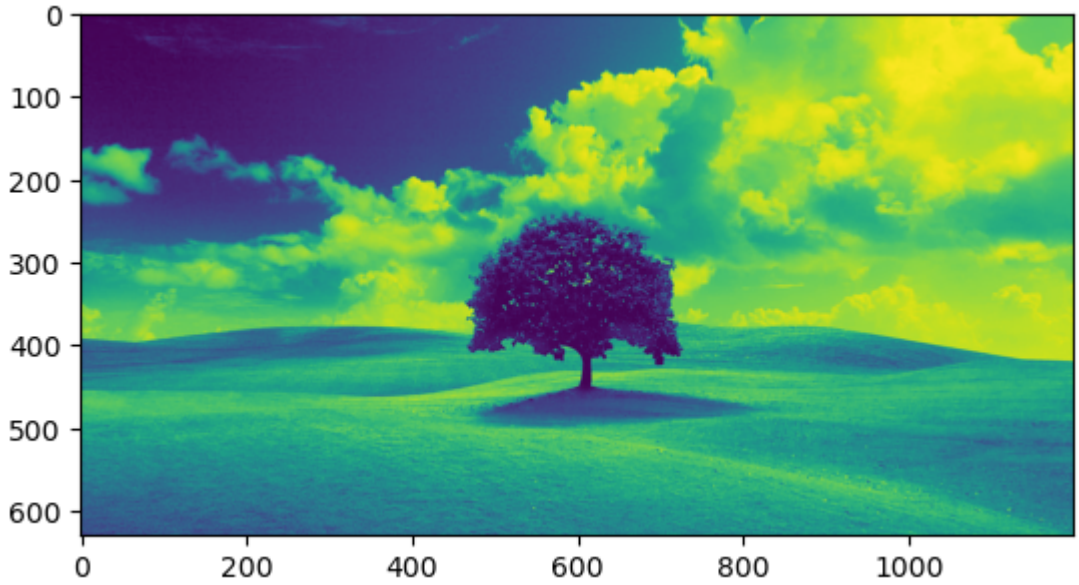


In [33]:
```
image_red.shape
```

Out[33]:
```
(630, 1200, 3)
```

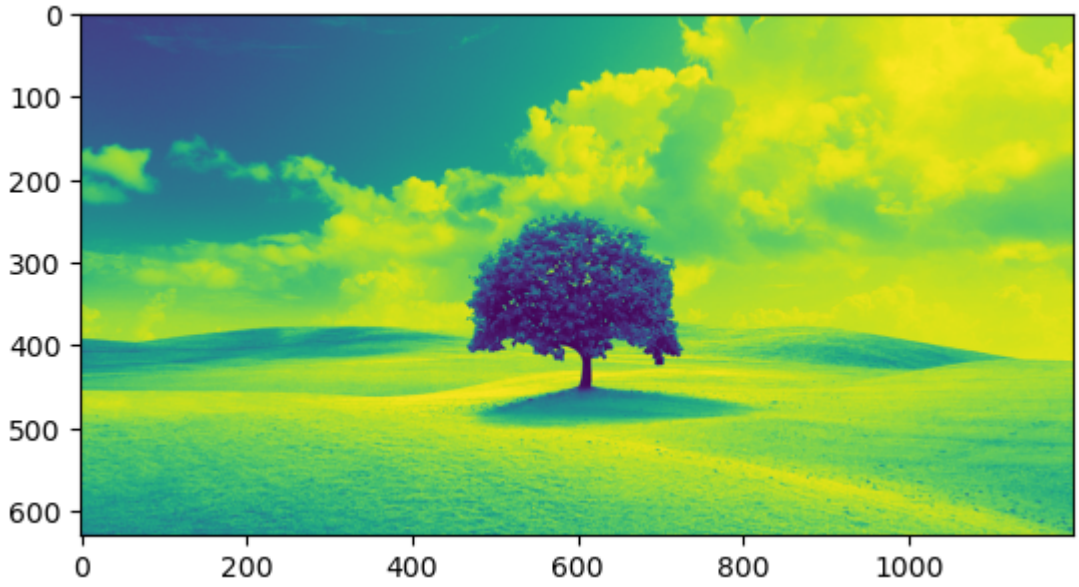
In [35]:
```
plt.imshow(image_red[:,:,0])
```

Out[35]:
```
<matplotlib.image.AxesImage at 0x21d0f8fe990>
```



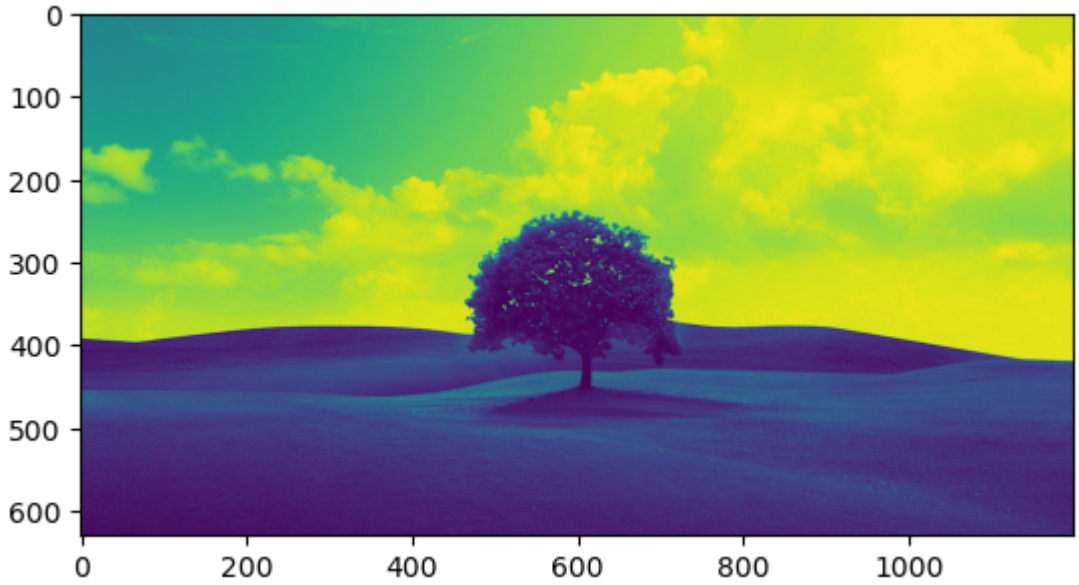
In [37]:
```
plt.imshow(image_red[:,:,1])
```

Out[37]:
```
<matplotlib.image.AxesImage at 0x21d0f7de710>
```



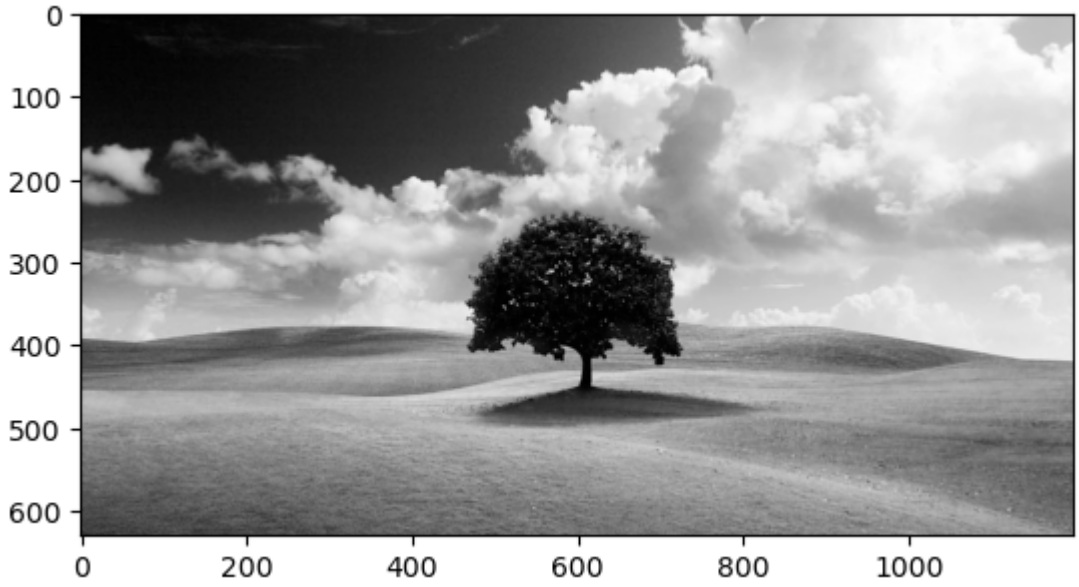
In [38]:
```
plt.imshow(image_red[:,:,2])   # R G B
```

Out[38]: <matplotlib.image.AxesImage at 0x21d0f830cd0>



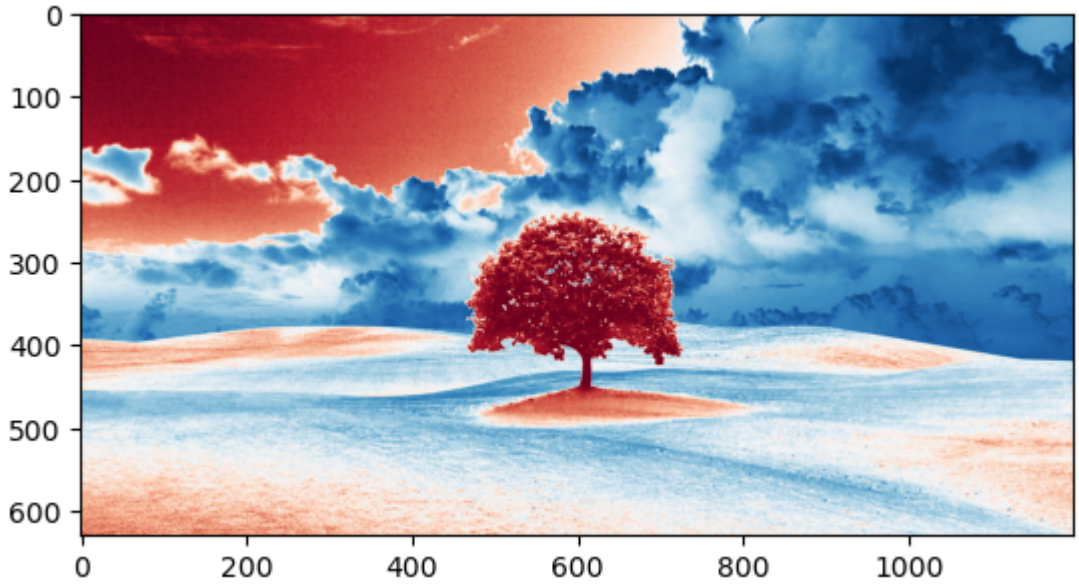
In [39]:
```
plt.imshow(image_red[:,:,0], cmap='gray')
```

Out[39]: <matplotlib.image.AxesImage at 0x21d102ab250>



In [46]:
```
plt.imshow(image_red[:,:,0], cmap='RdBu')
```

Out[46]:
```
<matplotlib.image.AxesImage at 0x21d12a9c690>
```



In [42]:
```
image_red[:,:,0]
```

Out[42]:
```
array([[  1,   2,   4, ..., 193, 194, 194],
       [  1,   2,   4, ..., 194, 194, 194],
       [  1,   2,   4, ..., 194, 194, 194],
       ...,
       [ 65,  68,  67, ..., 126, 121, 124],
       [ 65,  62,  60, ..., 139, 130, 133],
       [ 61,  54,  42, ..., 144, 135, 134]],
      shape=(630, 1200), dtype=uint8)
```

In [43]:
```
image_red[:,:,1]
```

Out[43]:
```
array([[ 49,  50,  49, ..., 220, 221, 221],
       [ 49,  50,  49, ..., 221, 221, 221],
       [ 49,  50,  49, ..., 221, 221, 221],
       ...,
       [123, 128, 124, ..., 194, 192, 198],
       [124, 119, 115, ..., 206, 199, 205],
       [120, 110,  96, ..., 209, 202, 202]],
      shape=(630, 1200), dtype=uint8)
```

In [44]:
```
image_red[:,:,2]
```

Out[44]:
```
array([[113, 114, 114, ..., 237, 238, 238],
       [113, 114, 114, ..., 238, 238, 238],
       [113, 114, 114, ..., 238, 238, 238],
       ...,
       [ 10,   6,  19, ...,  33,  26,  27],
       [  8,   2,  14, ...,  41,  30,  33],
       [  4,   0,   0, ...,  43,  34,  31]],
      shape=(630, 1200), dtype=uint8)
```

In [ ]:

In [ ]: